nonfunctional environment ensures that you not only get a grip on the infrastructure side of things but also understand the search and file I/O performance metrics in detail.

Official documentation, discussion forums (Stack Overflow), and blog posts for engineers are available to help you manage and monitor clusters. You can follow their guidance to tweak the memory configuration for optimal performance, disk usage, and smooth running of the cluster.

This chapter concludes our journey through learning, understanding, and working with Elasticsearch. Elasticsearch is a complex beast; it requires dedicated knowledge and committed expertise to maintain and run production farms. The good news is that the Elastic folks have produced rich documentation about the product over the years (although it's a bit dry and overwhelming sometimes), which can help you when you're lost—and of course, you can refer to this book!

Summary

- Elasticsearch is a complex search engine, and maintaining and managing healthy clusters requires expertise.
- Slower search speeds are a common problem that clients complain about when searching for data with Elasticsearch. Provisioning modern hardware with appropriate allocation of memory and computing resources helps alleviate speed problems. Options like choosing keyword data types and tweaking the document data model also help increase search speed.
- Indexing speeds are a concern as well, especially when the system is ingestion heavy. If allowed, measures like using system-generated IDs for documents can speed up the indexing process. Loading data using bulk APIs (rather than single-document APIs) is a surefire way to improve ingestion performance.
- We can turn off or increase the refresh rate during indexing and then switch it on when indexing completes. Doing so makes the documents unavailable instantly for searches but improves indexing performance by reducing I/O hits.
- Due to the distributed nature of Elasticsearch, things can go wrong in many areas, from the cluster to the filesystem, node communication, memory, and so on. Keeping a healthy cluster is paramount, and administrators must strive to maintain a GREEN cluster state. Religiously following the traffic light management-based cluster health system helps with smooth running of the cluster.
- Occasionally, shards are left unassigned, and troubleshooting to find the exact cause helps us reattach nodes if necessary or take the appropriate action.
- Elasticsearch provides thresholds on the available disk memory, including low-, high-, and flood-stage-disk watermarks. The low- and high-disk watermarks allow Elasticsearch to manage shard reallocation and alert the administrator about upcoming cluster problems. The flood-stage watermark is a dire warning